Name: ___________________________

Homework Questions for Climate and Us Week 1
Due the week of November 17
SHOW ALL WORK

1. Earth's Energy Balance

Any given planet's climate is determined by its energy balance. Earth's energy balance is such that it creates a livable environment for humans and other life, unlike those of nearby Venus or Mars. Climate forcings are factors that drive changes in climate and alter the Earth's energy balance. **Use Figure 1 (below) to answer the following questions.**

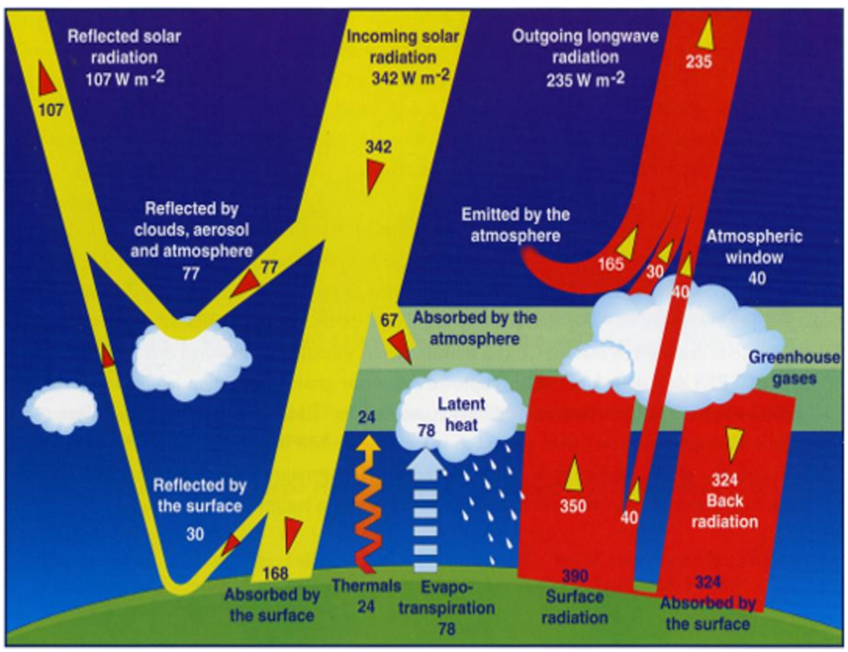


Figure 1. *Energy balance diagram showing the flow of energy (in Watts per square meter, or W/m^2) from the Sun, from Earth's surface, and through the atmosphere. Yellow regions show incoming solar (shortwave) radiation. Red regions show outgoing infrared (longwave) radiation.*

1A. (i) What three values illustrate the balance between incoming and outgoing energy? Set up an equation with these three values illustrating this balance.

Incoming solar at top of atmosphere: 342 W/m²
Reflected solar: 107 W/m²
Outgoing longwave: 235 W/m²

Energy balance equation:
342 = 107 + 235

1A. (ii) Where does the incoming solar radiation go? Name the possible outcomes for incoming solar radiation based on the diagram.

Solar radiation it can:

Be reflected by clouds and atmosphere

Be reflected by Earths surface

Be absorbed by the atmosphere

Be absorbed by Earths surface

1A. (iii) What values illustrate the balance between energy absorbed at the surface and lost at the surface? Select all that apply and demonstrate the two energy flows are equal with a calculation.

Energy absorbed at the surface:

a. Solar radiation: 168 W/m^2
b. Thermals: 24 W/m^2
c. Surface radiation: 390 W/m^2
d. Back radiation: 324 W/m^2
e. Evapotranspiration: 78 W/m^2

Energy lost at the surface:

a. Solar radiation: 168 W/m^2
b. Thermals: 24 W/m^2
c. Surface radiation: 390 W/m^2
d. Back radiation: 324 W/m^2
e. Evapotranspiration: 78 W/m^2

Total absorbed: 168 + 324 = 492 W/m2 and Total lost: 24 + 390 + 78 = 492 W/m2 so they equal

1A. (iv) The outgoing longwave radiation is made up of energy emitted by the atmosphere (165 W/m^2) plus energy emitted by the clouds (30 W/m^2), plus energy escaping through the atmospheric window (40 W/m^2).

What components together account for the energy emitted by the atmosphere plus that emitted by the clouds? Select all that apply. *Hint: think about which components need to be added OR subtracted.*

a. Energy absorbed by the atmosphere coming in: 67 W/m^2
b. Thermal energy to the atmosphere: 24 W/m^2
c. Evapotranspiration to make clouds: 78 W/m^2
d. Energy reflected by the surface: 30 W/m^2
e. Surface radiation (not including atmospheric window): 350 W/m^2
f. Surface radiation (including atmospheric window): 390 W/m^2
g. Back radiation: -324 W/m^2

1B. (i) How would an increase in albedo, defined as the amount of shortwave energy reflected by the Earth (surface + atmosphere), affect the energy balance in Figure 1? Write 'increase', 'decrease', or 'not change' in the provided blanks.

a. Shortwave energy absorbed by Earth's surface (168 W/m^2) would decrease
b. Longwave energy input to the atmosphere by surface radiation (350 W/m^2) would decrease
c. Incoming solar radiation (342 W/m^2) would n't change
d. Earth's average surface temperature would decrease

1B. (ii) Table 1 below gives the value of albedo for various surfaces. **Which of the following would cause a decrease in Earth's surface albedo?** Select all that apply.

Substance or Surface	Albedo
Whole Earth average	0.3
Fresh snow	0.8-0.9
White paint	0.5-0.9
Sea ice	0.5-0.7
Desert sand	0.4
Green grass	0.25
Bare soil	0.17
Dark roofs	0.1-0.2
Conifer forest	0.08-0.15
Open ocean	0.06
Fresh asphalt	0.04

Table 1. *Albedo of various substances and surfaces, defined as the fraction of solar radiation reflected. (Data from Akbari et al., 2012)*

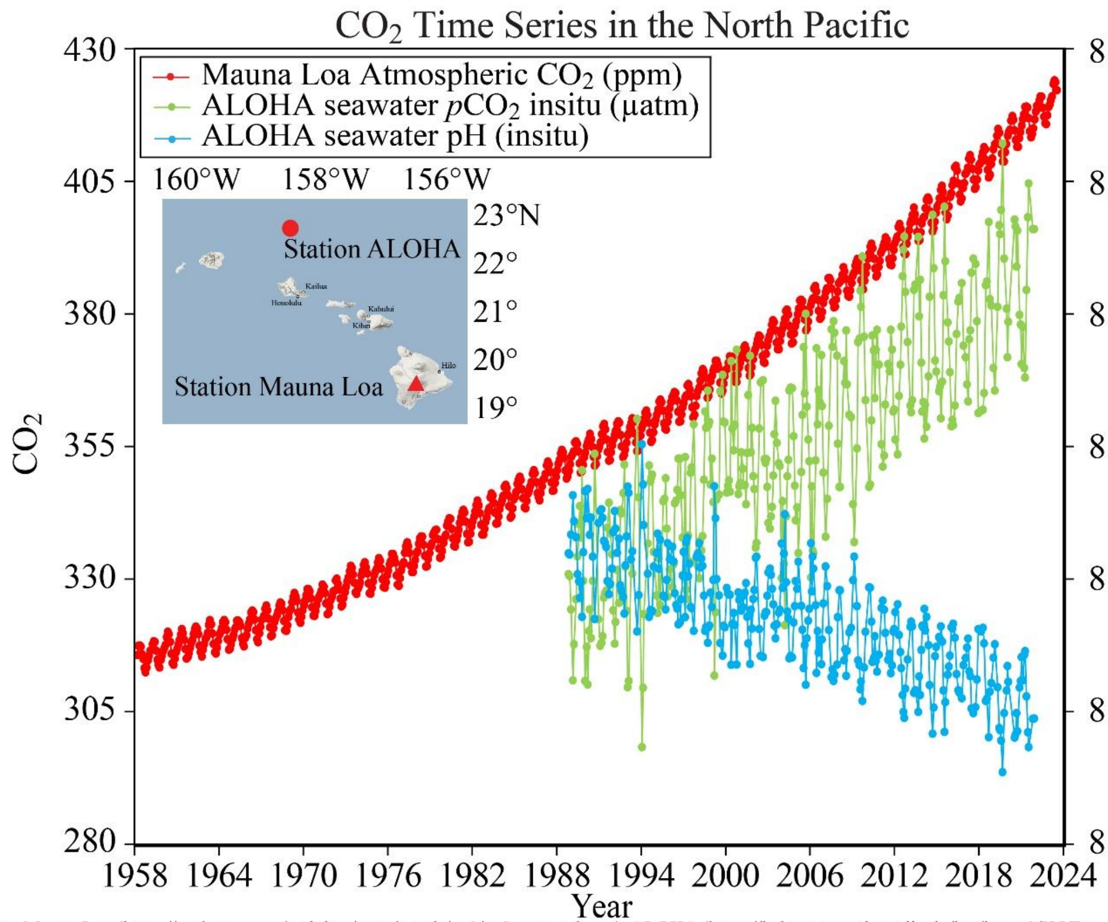
a. Allowing the size of the Sahara Dessert to increase and take over forests
b. Melting sea ice
c. Replacing a grassy field or forested park with a parking lot
d. Planting grass in open fields of bare soil
e. Sea level rise increasing the surface area of Earth's oceans
f. Painting roofs of buildings white

1C. TRUE OR FALSE (correct the statement if false): Greenhouse gases largely absorb incoming solar (shortwave) radiation.

It's false, greenhouse gases absorb outgoing infrared longwave radiation

2. <u>Recent Climate Change and Impacts</u>

Figure 2 (next page) shows measurements of atmospheric CO_2 from the Mauna Loa Observatory station in Hawaii starting in 1958. Seawater pH and seawater partial pressure CO_2 have also been measured at the ALOHA station off the coast of the Hawaiian island O'ahu (see inset map) since approximately 1990, as shown in Figure 2.



Data: Mauna Loa (https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt) ALOHA (https://hahana.soest.hawaii.edu/hot/hotco2/HOT_surface_CO2.txt)
ALOHA pH & pCO_2 are calculated at in-situ temperature from DIC & TA (measured from samples collected on Hawaii Ocean Times-series (HOT) cruises)
using co2sys (Pelletier, v25b06) with constants: Lueker et al. 2000, KSO4: Dickson, Total boron: Lee et al. 2010, & KF: seacarb

Figure 2. *Measurements of atmospheric CO_2 concentration (red line) from the Mauna Loa Observatory station from 1958 to 2024, seawater partial pressure of CO_2 in μatm (green line) and seawater pH (blue line) from station ALOHA from 1990 to February 2024. Source: National Oceanic and Atmospheric Administration Pacific Marine Environmental Laboratory Carbon Program.*

2A. The record of atmospheric CO_2 concentrations from Mauna Loa, Hawaii shows both a long-term trend from 1958-2024, as well as short-term (intra-annual) variability. **Explain what causes intra-annual variability in atmospheric CO_2 in 2-3 sentences.**

Intra annual variability in atmospheric CO2 actually comes from the seasonal cycle of plant growth and death. In the growing season, plants take up more CO2, so atmospheric CO2 is lower. In the other season, respiration and death release CO2 back to the air, so atmospheric CO2 actually rises

2B. (i) Using Figure 2, estimate the annual rate of change for atmospheric CO_2 concentration (in ppm CO_2 per year) at the Mauna Loa Observatory station for the time period from 1958-1976. Show your work, then select the multiple choice answer that is closest to the value you calculated.

a. 0.05 ppm/year
b. 0.1 ppm/year
c. 0.8 ppm/year
d. 1.6 ppm/year
e. 2.3 ppm/year

Change is around 332 − 315 = 17 ppm over 18 years

Rate is around 17 ÷ 18 = 0.94 ppm per year

2B. (ii) Now estimate the average annual rate of change for the time period from 2006 to 2024 (Column A). How many times larger is this annual rate of change in atmospheric CO_2 compared to the period from 1958-1976, roughly half a century earlier (Column B)? Show your work, then select the multiple choice answers that are closest to the values you calculated. Note: use the multiple choice value selected in Column A to determine the rate for Column B.

A	B
a. 0.05 ppm/year	f. It is the same rate
b. 0.1 ppm/year	g. The 2006-2024 rate is 1.5 times larger
c. 0.8 ppm/year	h. The 2006-2024 rate is 3 times larger
d. 1.6 ppm/year	i. The 2006-2024 rate is 4 times larger
e. 2.3 ppm/year	j. The 2006-2024 rate is 5 times larger

Change is around 420 − 380 = 40 ppm over 18 years

Rate is around 40 ÷ 18 ≈ 2.2 ppm per year

then we compare the 1958 to 1976 rate (ratio is 2.3 ÷ 0.8 = 2.9)

2C. As the concentration of CO_2 in the atmosphere increases, the concentration of CO_2 dissolved in seawater also increases, causing a change in pH. **Based on Figure 2, what can we conclude about the relationship between atmospheric CO_2 concentration and seawater acidity?** Note that a decrease in pH means the water is becoming more acidic while an increase in pH means the water is becoming more basic.

As atmospheric CO2 increases, seawater pCO2 also increases, while seawater pH decreases. Lower pH means higher acidity so higher atmospheric CO2 goes together with more acidic ocean water

2D. The ocean is a major carbon "sink" and currently absorbs about 31% of the CO_2 released by anthropogenic sources into the atmosphere (Gruber et al., 2019). This means the rate you calculated in question 2B is actually only 69% of what it would be if the ocean were not able to absorb atmospheric CO_2. Unfortunately, the ocean's ability to absorb CO_2 is finite, and has been declining over the past decade as warmer oceans have less capacity for CO_2 uptake. **Based on your *multiple choice answer* to 2B (ii), what would the concentration of atmospheric CO_2 be this year (2025) if ocean absorption of atmospheric CO_2 had all of a sudden gone from 31% per year to 0% per year starting in 2006 (when atmospheric CO_2 was 380 ppm)?** Compare this value to the observed atmospheric CO_2 that we have today of 425 ppm (Oct 2025 average).

rate without ocean uptake:
2.3 ÷ 0.69 = 3.3 ppm per year

2006 to 2025 is 19 years:
Increase = 3.3 × 19 = 63 ppm

Starting from 380 ppm in 2006:
380 + 63 = 443 ppm

without ocean absorption, 2025 atmospheric CO2 would be a little around 4.4 × 10^2 ppm, instead of the 425 ppm. That's about 18 ppm higher

2E. Figure 3 shows the flux of carbon (in petagrams of C per year) going into the atmosphere measured at the Mauna Loa Observatory in Hawaii (blue line), in ice core bubbles from Antarctica (black line) and released by the burning of fossil fuels (red line).

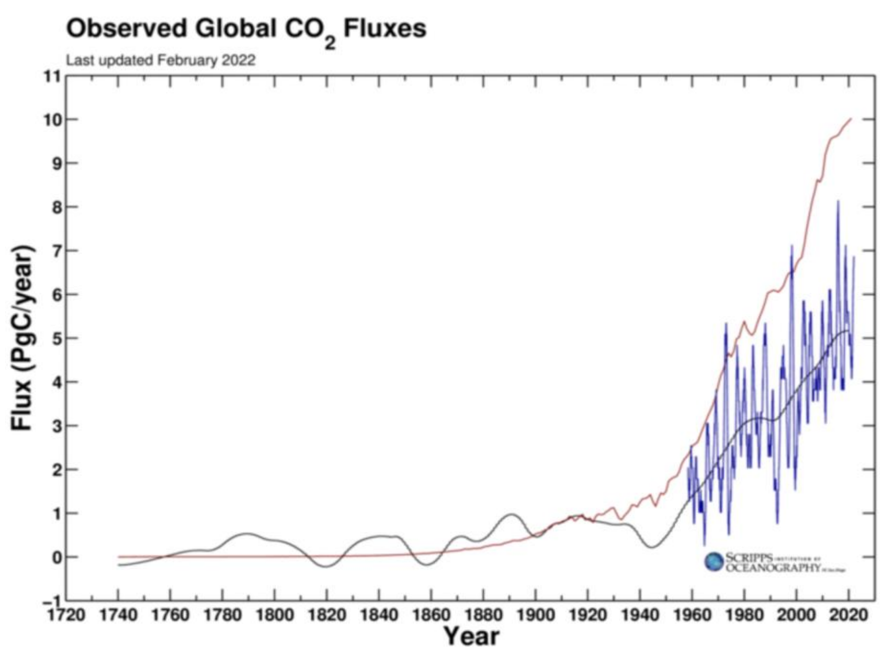


Figure 3. *The annual flux of CO_2 (mass per year) is calculated using the yearly increase in atmospheric CO_2 concentration. Data from Scripps Institute of Oceanography.*

2E. (i) Based on Figure 3 (previous page), estimate the difference between the flux of carbon released by fossil fuels and the flux of carbon measured in Antarctica in 2020. Give your answer in kilograms of carbon per day and write your answer in scientific notation (1 petagram = 1 x 10^{15} grams).

Difference in flux = 10 – 5 = 5 Pg C per year

Convert to kilograms of carbon per day:

1 Pg = 10^15 g = 10^12 kg

5 Pg C per year = 5 × 10^12 kg C per year

per day: (5 × 10^12 kg C per year) / 365 = 1.4 × 10^10 kg c per day

2E. (ii) We already learned that some of the carbon released by the burning of fossil fuels is being absorbed by the ocean. Where else could it be going? **Provide another reasonable carbon sink on Earth where the excess carbon you calculated in 2E. (i) is going.**

Another major sink for this missing carbon is the land biosphere, since extra CO2 is taken up by the big vegetation

2F. In Prof. Helfand's lecture, you learned about the use of isotope ratios to discern the source of atmospheric carbon. Figure 4 below shows the ratio of $^{13}C/^{12}C$ in atmospheric CO_2 (relative to a standard), which is denoted as $\delta^{13}C$. The black line shows the value recorded at the Mauna Loa Observatory, Hawaii, and the red line shows the value recorded at the South Pole, Antarctica.

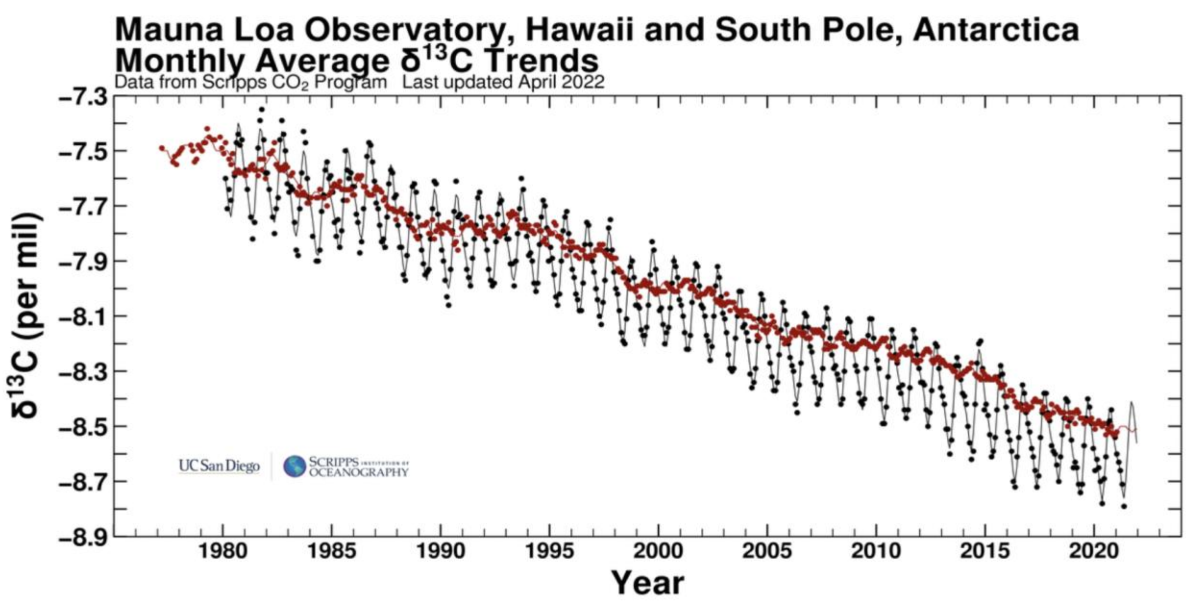


Figure 4. *Average monthly $\delta^{13}C$ records of atmospheric CO_2 from Mauna Loa Observatory, Hawaii (black dots/curve) and South Pole, Antarctica (red dots/curve) from 1975 to 2022. (Note: $\delta^{13}C$ represents the ratio of $^{13}C/^{12}C$ relative to a standard and is recorded in parts per thousand or "per mil"). Source: Scripps CO_2 program.*

2F. (i) Changes in $\delta^{13}C$ represent the ratio of $^{13}C/^{12}C$ relative to a standard, so a higher (less negative) $\delta^{13}C$ corresponds to a higher $^{13}C/^{12}C$ ratio and a lower (more negative) $\delta^{13}C$ corresponds to a lower $^{13}C/^{12}C$ ratio. **Based on what you learned in lecture, why does a decline in $\delta^{13}C$ through time provide support that fossil fuels are the source of increasing atmospheric CO_2?** Explain in 2-3 sentences.

Fossil fuels come from old organic matter that's low in carbon 13 and when we burn these fuels, we add carbon dioxide that has a low 13C ratio to the air

figure 4 shows that 13C in atmospheric CO2 became a little negative at both Mauna Loa and the South Pole from the 1970s to today. A more negative 13C means the air is gaining carbon with less 13C, which is pretty much what we expect from fossil fuel carbon and not from volcanoes or the ocean.
so the long term drop in 13C supports the conclusion that the rising atmospheric CO2 mainly comes from burning fossil fuels

2F. (ii) Which location shows greater intra-annual variability in $\delta^{13}C$ value of atmospheric carbon dioxide: the South Pole, Antarctica or Mauna Loa, Hawaii? Circle the correct answer and in 2-3 sentences, speculate as to what may be responsible for this difference.

in the figure the Mauna Loa has sharp ups and downs each year while the South Pole is much smoother with smaller swings

Mauna loa is closer to sinks of CO2 especially pollution in the Northern Hemisphere so these nearby factors cause a lot of seasonal changes in both Co2 and the isotopic composition

South pole is far from large land areas and pollution so air there is more mixed from all directions so the seasonal variations in 13C are weaker